

Causal Inference

Lecture 11

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Based partially on Prof. Arvid Sjölander's Mediation analysis lecture slides

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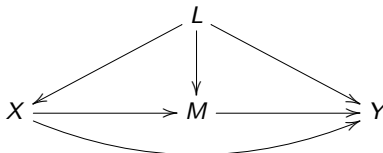


- 1 Motivating example
- 2 Controlled direct effect
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- 4 Identification and Estimation
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Notations

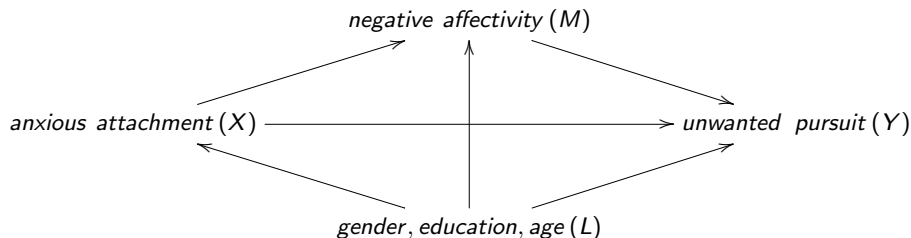
- Let X , M , and Y , the exposure, mediator and outcome, respectively.
- Let L be a set of measured covariates that we wish to condition on in the analysis.
- We will assume the following causal structure (no unmeasured confounders)



DAG for mediation analysis.

Negative affectivity mediates the effect of attachment style on pursuit behavior

- De Smet et al. (2012) and Loeys et al. (2013) proposed emotional distress or the amount of negative affectivity experienced during the breakup as a mediating variable for the effect of attachment style towards the ex-partner before the breakup on displaying unwanted pursuit behaviors after the breakup.
- Subsample of 385 subjects publicly available in R package medflex (Steen et al., 2017)



Negative affectivity experienced during the breakup as a mediating variable for the effect of attachment style on unwanted pursuit behavior

Variables

- att: self-reported anxious attachment level before divorce
- attbin: dichotomized att (0/1 for below/above mean) - **exposure** X
- attcat: categorized att (L=low, M=middle, H=high)
- negaff: level of self-reported experienced negative affectivity (e.g. emotional distress) during divorce - **mediator** M
- initiator: initiator of the divorce
- gender: gender (F=female, M=male)
- educ: education level (H=high, M=middle, L=low)
- age: age (years)
- UPB: indicator of displayed unwanted pursuit behavior towards the ex-partner after divorce - **outcome** Y
- The study analyze repeated and unwanted efforts to establish intimate contact in the form of harassing, tracking, and monitoring. (Foshay and O'Sullivan, 2019)

Logistic regression

- To estimate the total effect of anxious attachment level on UPB we may use logistic regression

```
Call:
glm(formula = UPB ~ attbin + gender + educ + age, family = binomial,
    data = UPBdata)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.95690    0.68983  -1.387  0.165396
attbin       0.74283    0.21830   3.403  0.000667 ***
genderM      0.29426    0.22504   1.308  0.191016
educM        0.36973    0.47484   0.779  0.436192
educH        0.54075    0.48193   1.122  0.261841
age          -0.01200    0.01188  -1.010  0.312310
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Logistic regression fit for the total effect of anxious attachment level on pursuit behavior

Questions we strive to answer the end of this lecture

- What is the direct effect of anxious attachment level on UPB, not mediated through negative affectivity?
- What is the indirect effect of anxious attachment level on UPB, mediated through negative affectivity?
- What proportion of the total effect is mediated?

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Notations

- Let $Y_{x,m}$ be the potential outcome of Y when X and M are set to x and m , respectively.
- Let g be the link function, the conditional CDE at $M = m$, given X , is defined as

$$CDE(m, L) = g(\mathbb{P}(Y_{1,m} = 1|L)) - g(\mathbb{P}(Y_{0,m} = 1|L))$$

- For example, Identity link function, risk ratio

$$\mathbb{P}(Y_{1,m} = 1|L) - \mathbb{P}(Y_{0,m} = 1|L)$$

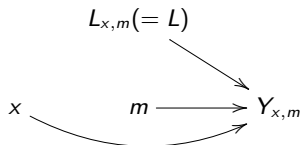
- log risk ratio

$$\log(\mathbb{P}(Y_{1,m} = 1|L)) - \log(\mathbb{P}(Y_{0,m} = 1|L))$$

- log odds (logit) ratio

$$\log\left(\frac{\mathbb{P}(Y_{1,m} = 1|L)}{\mathbb{P}(Y_{1,m} = 0|L)}\right) - \log\left(\frac{\mathbb{P}(Y_{0,m} = 1|L)}{\mathbb{P}(Y_{0,m} = 0|L)}\right)$$

$$\text{CDE}(m, L) = g\{\mathbb{P}(Y_{1,m} = 1 \mid L)\} - g\{\mathbb{P}(Y_{0,m} = 1 \mid L)\}.$$

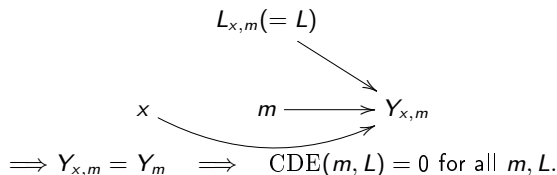


Remarks

- Setting M to m eliminates the arrow $X \rightarrow M$ (and the arrow $L \rightarrow M$)
- Any remaining effect of X on Y must be *direct*.
- By definition, $\text{CDE}(m)$ is the average causal effect on the g -scale of the treatment if the intermediate variable is fixed at m .
- The parameter $\text{CDE}(m)$ can capture the direct effect of the treatment holding the mediator at m . However, this formulation cannot capture the indirect effect.
- In particular, the parameter $\mathbb{E}[Y_{x,1} - Y_{x,0}]$ only measures the effect of the mediator on the outcome holding the treatment at x . This is not a meaningful definition of the indirect effect.

$$\text{CDE}(m, L) = g\{\mathbb{P}(Y_{1m} = 1 \mid L)\} - g\{\mathbb{P}(Y_{0m} = 1 \mid L)\}.$$

- $\text{CDE}(m, L)$ generally depends on both m and L .
- Exception (no direct effect):



If $Y_{x,m} = Y_m$ for all m, L , then $\text{CDE}(m, L) = 0$.

Numerical toy example

$$CDE(m, L) = g\{\mathbb{P}(Y_{1m} = 1 \mid L)\} - g\{\mathbb{P}(Y_{0m} = 1 \mid L)\}.$$

ID	Y_{00}	Y_{01}	Y_{10}	Y_{11}
1	0	0	1	1
2	0	1	1	1
3	0	0	0	1

- For the data above, compute $CDE(0)$ and $CDE(1)$ as risk differences.

Numerical toy example - solution

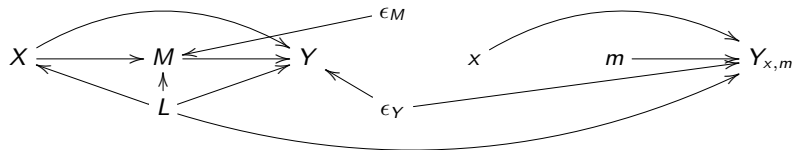
$$\text{CDE}(m, L) = g\{\mathbb{P}(Y_{1m} = 1 \mid L)\} - g\{\mathbb{P}(Y_{0m} = 1 \mid L)\}.$$

ID	Y_{00}	Y_{01}	Y_{10}	Y_{11}
1	0	0	1	1
2	0	1	1	1
3	0	0	0	1

$$\text{CDE}(0) = \mathbb{P}(Y_{10} = 1) - \mathbb{P}(Y_{00} = 1) = \frac{2}{3} - 0 = \frac{2}{3}.$$

$$\text{CDE}(1) = \mathbb{P}(Y_{11} = 1) - \mathbb{P}(Y_{01} = 1) = 1 - \frac{1}{3} = \frac{2}{3}.$$

No effect modification by M for the controlled direct effect, on the risk difference scale.



Observed studies and Experiments

- Conditional exchangeability (observed study where no randomization is assumed)

$$Y_{x,m} \perp\!\!\!\perp (X, M) \mid L$$

- From potential outcome to observed data

$$\mathbb{E}[Y_{x,m} \mid L] = \mathbb{E}[Y_{x,m} \mid X = x, M = m, L] = \mathbb{E}[Y \mid X = x, M = m, L].$$

When L is high-dimensional we could use a regression to model $\mathbb{E}[Y_{x,m} \mid L; \beta]$.

- Notably, an unconditional exchangeability $Y_{x,m} \perp\!\!\!\perp (X, M)$, can be attained by using randomized trial where **both the exposure and the mediator are randomized**. In such a case, the CDE is identifiable even without using any covariates L .

Disadvantages

- Has no explicit analog for indirect effect
- Can define indirect effect (IE) as total effect (TE) minus the direct effect (DE)

$$IE = TE - DE$$

but this is implicit definition is somewhat ad hoc.

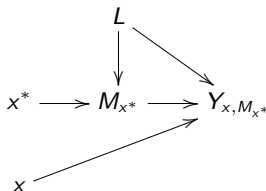
- An explicit definition can be constructed by using 'natural effects' instead

Advantages

- The formulation of the controlled direct effect does not involve nested potential outcomes, and its identification does not require the cross-world counterfactual independence assumption.
- Identifiable in randomized trials without additional unverifiable assumption.
- Assuming a linear outcome model without exposure-mediator interaction, the CDE simplifies to the coefficient of the treatment, i.e., the same as the NDE.
- Since $TE = NDE + NIE$, under linear models, the Baron-Kenny computation method of the indirect effect is also consistent with the NIE. Hence, the CDE is easy to understand and easy to estimate.

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$$\text{NDE}(x^*, L) = g\{\mathbb{P}(Y_{1,M_{x^*}} = 1 \mid L)\} - g\{\mathbb{P}(Y_{0,M_{x^*}} = 1 \mid L)\}.$$

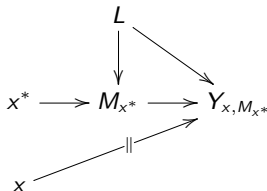


DAG that illustrates the NDE using the effect of two different worlds.

NDE - interpretation

- By holding x^* fixed, we ensure that the mediator does not vary *due to* the exposure.
 - It may still vary for other reasons (e.g. because L changes).
- Hence, any variation in Y when x is taken from 0 to 1 must be due to a *direct* effect of the exposure.

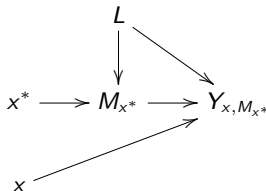
$$\text{NDE}(x^*, L) = g\{\mathbb{P}(Y_{1,M_{x^*}} = 1 \mid L)\} - g\{\mathbb{P}(Y_{0,M_{x^*}} = 1 \mid L)\}.$$



DAG that illustrates the NDE using the effect of two different worlds.

- $\text{NDE}(x^*, L)$ generally depends on both x^* and L , unless there is no direct effect. In such a case
 - $Y_{x,M_{x^*}} \equiv Y_{M_{x^*}}$, thus $\text{NDE}(x^*, L) = 0 \quad \forall x^*, L$.

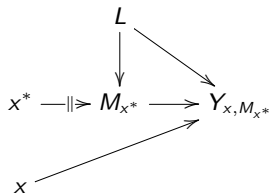
$$\text{NIE}(x, L) = g\{\mathbb{P}(Y_{x, M_1} = 1 \mid L)\} - g\{\mathbb{P}(Y_{x, M_0} = 1 \mid L)\}.$$



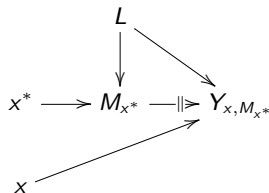
NIE - interpretation

- By holding x fixed we ensure that the outcome does not vary due to a direct effect of the exposure.
- By taking x^* from 0 to 1 we let the mediator vary due to the exposure, and this variation is propagated to the outcome.
- Hence, any variation in the outcome when x^* is taken from 0 to 1 must be due to an indirect effect of the exposure.

No indirect effect



(a) $Y_{x, M_{x^*}} = Y_{x, M}$



(b) $Y_{x, M_{x^*}} = Y_x$

$$NIE(x, L) = 0, \quad \forall x, L$$

No indirect effect

If there is no causal effect of the exposure x^* on the mediator M_{x^*} or from the mediator M_{x^*} on the potential outcome $Y_{x, M_{x^*}}$, then there is no (natural) indirect effect of the exposure on the outcome.

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Sequential ignorability

The identification of the natural direct and indirect effects relies on sequential ignorability, i.e.,

$$\{Y_{x^*,m}, M_x\} \perp\!\!\!\perp X | L$$

which means no unmeasured confounders between the exposure and the outcome and between mediator and the exposure. Additionally, we require

$$Y_{x^*,m} \perp\!\!\!\perp M_x | X = x, L$$

which is the cross-world independence, and implies no unmeasured confounders between the mediator and the outcome.

The mediation formula

$$\mathbb{E}[Y_{x,M_{x^*}} | L] = \mathbb{E}[\mathbb{E}[Y | X = x, M_{x^*} = m, L] | X = x^*, L]$$

- We can fit regression model to $\mathbb{E}[Y|X = x, M_{x^*} = m, L; \hat{\beta}]$ and $\mathbb{P}(M|X = x^*, L; \hat{\gamma})$, and then estimate the nested cross-world counterfactuals mean as (using mediation formula)

$$\mathbb{E}[Y_{x, M_{x^*}} | L] = \sum_m \mathbb{E}[Y|X = x, M_{x^*} = m, L; \hat{\beta}] \mathbb{P}(M = m | X = x^*, L; \hat{\gamma}).$$

- The vector of estimands is $\theta = (\beta^T, \gamma^T, \mathbb{E}[Y_{0, M_0} | L], \mathbb{E}[Y_{0, M_1} | L], \mathbb{E}[Y_{1, M_0} | L], \mathbb{E}[Y_{1, M_1} | L])^T$.
- The estimators solve the following systems of equations simultaneously

$$\sum_{i=1}^n S_i(\beta) = 0, \quad \sum_{i=1}^n S_i(\gamma) = 0, \quad \sum_{i=1}^n H_i(\theta) = 0,$$

where $S_i(\beta)$ and $S_i(\gamma)$ are unbiased estimating functions for β and γ , respectively (e.g. ML score functions), and $H(\theta)$ is the estimating function for the causal estimands

$$H_i(\theta) = \begin{cases} \sum_m \mathbb{E}[Y | X = 0, M = m, L_i; \beta] \mathbb{P}(M = m | X = 0, L_i; \gamma) - \mathbb{E}[Y_{0, M_0} | L_i], \\ \sum_m \mathbb{E}[Y | X = 0, M = m, L_i; \beta] \mathbb{P}(M = m | X = 1, L_i; \gamma) - \mathbb{E}[Y_{0, M_1} | L_i], \\ \sum_m \mathbb{E}[Y | X = 1, M = m, L_i; \beta] \mathbb{P}(M = m | X = 0, L_i; \gamma) - \mathbb{E}[Y_{1, M_0} | L_i], \\ \sum_m \mathbb{E}[Y | X = 1, M = m, L_i; \beta] \mathbb{P}(M = m | X = 1, L_i; \gamma) - \mathbb{E}[Y_{1, M_1} | L_i]. \end{cases}$$

Complexity and communication challenge

- The estimating equations for the causal estimands are derived from the mediation formula

$$\mathbb{E}[Y_{x, M_{x^*}} \mid L] = \sum_m \mathbb{E}[Y \mid X = x, M = m, L; \beta] \mathbb{P}(M = m \mid X = x^*, L; \gamma).$$

- The natural direct and indirect effects are complex functions of (β, γ, L)
 - In particular, the effects generally depend on L , even if the models have no interaction terms.
- Hard to communicate the result.
- Awkward to test the null hypothesis of no direct or indirect effect.
- The analytical variance is complicated for computation due to the complex functions of $(\beta^T, \gamma^T, L)^T$ that the estimating equations involve.

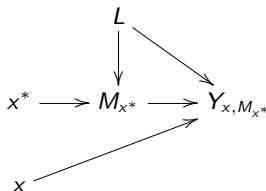
Structural mean models for the nested cross-world potential outcomes

- The problem can be solved by using an explicit structural mean model for the nested potential cross-world outcome

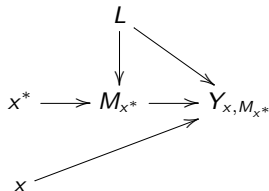
$$\mathbb{E}[Y_{x,M_{x^*}} \mid L; \psi].$$

- For example,

$$\text{logit}\{\mathbb{P}(Y_{x,M_{x^*}} = 1 \mid L; \psi)\} = \psi_3 + \psi_0 x + \psi_1 x^* + \psi_2 L.$$



Structural mean models for the nested cross-world potential outcomes



- The structural mean model for the potential cross-world nested outcome

$$\text{logit}\{\mathbb{P}(Y_{x, M_{x^*}} = 1 \mid L; \psi)\} = \psi_3 + \psi_0 x + \psi_1 x^* + \psi_2 L.$$

- All $\psi_0, \dots, \psi_3$ are causal parameters, however, ψ_3 and ψ_2 are nuisance parameters. Namely, although, these parameters are estimated, we are interested only in ψ_0 and ψ_1 .
- Although the potential mediator M_{x^*} has a direct effect on the potential outcome, if we ignore it in the structural model, thus its effect is incorporated in the coefficient of x^* , ψ_1 , that accounts for the causal path $x^* \rightarrow M_{x^*} \rightarrow Y_{x, M_{x^*}}$, i.e., the NIE.
- The SMM (and thus the causal parameters) is a latent model since it depends on two different worlds effects (i.e., by the definition not observable quantities in any real-world setting).

Alternative: Explicit Model for Natural Effects

- The structural mean model for the nested counterfactual is

$$\text{logit}\{\mathbb{P}(Y_{x,M_{x^*}} = 1 \mid L; \psi)\} = \psi_3 + \psi_0 x + \psi_1 x^* + \psi_2 L.$$

- The natural effects are

$$\begin{aligned}\text{NDE}(x^*, L) &= \text{logit}\{\mathbb{P}(Y_{1,M_{x^*}} = 1 \mid L; \psi)\} - \text{logit}\{\mathbb{P}(Y_{0,M_{x^*}} = 1 \mid L; \psi)\} \\ &= \psi_3 + \psi_0 \times 1 + \psi_1 x^* + \psi_2 L - (\psi_3 + \psi_0 \times 0 + \psi_1 x^* + \psi_2 L) \\ &= \psi_0.\end{aligned}$$

$$\begin{aligned}\text{NIE}(x, L) &= \text{logit}\{\mathbb{P}(Y_{x,M_1} = 1 \mid L; \psi)\} - \text{logit}\{\mathbb{P}(Y_{x,M_0} = 1 \mid L; \psi)\} \\ &= \psi_3 + \psi_0 \times x + \psi_1 \times 1 + \psi_2 L - (\psi_3 + \psi_0 \times 0 + \psi_1 \times 0 + \psi_2 L) \\ &= \psi_1.\end{aligned}$$

- The model assumes that $\text{NDE}(x^*, L)$ and $\text{NIE}(x, L)$ are constant across L , and that there is no interaction between x and x^* .
- The models allow for a straightforward methods to test the null hypothesis of no indirect natural effect.
- The models allow to construct an analogical settings to the linear models, that are much more intuitive for interpretation.

Regression models for the natural effects

- There are two ways to fit the natural effects model:
 - **Imputation-based method** (Vansteelandt et al., 2012), which requires a nuisance model

$$\mathbb{E}[Y \mid X, M, L; \beta].$$

Roughly speaking, corresponds to the idea of G-computation (standardization).

- **Weighting-based method** (Lange et al., 2012), which requires a nuisance model

$$\mathbb{P}(M = m \mid X, L; \gamma).$$

Roughly speaking, corresponds to the idea of IPW.

- Note: there are version of doubly-robust estimator of direct effect (e.g., van der Laan, M. J., & Petersen, M. L., 2008), but these methods are outside the scope of this course.

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Estimating Equations

The parameters are

$$\theta = (\beta^T, \psi^T)^T.$$

The imputation-based estimator solves

$$\sum_{i=1}^n \begin{pmatrix} S_i(\beta) \\ H_i(\theta) \end{pmatrix} = 0,$$

where

$$H_i(\theta) = H_{i1}(\theta) + H_{i2}(\theta).$$

Components of $H_i(\theta)$

$$H_{i1}(\theta) = d(x_i, L_i) \left\{ Y_i - \mathbb{E}[Y_{x_i, M_{x_i}^*} \mid L_i; \psi] \right\},$$

$$H_{i2}(\theta) = d(x_i, L_i) \left\{ \mathbb{E}[Y \mid X = 1 - x_i, M_i, L_i; \beta] - \mathbb{E}[Y_{1-x_i, M_{x_i}} \mid L_i; \psi] \right\},$$

and $d(x_i, L_i)$ has the same dimension as ψ .

Definition

$$H_i(\theta) = H_{i1}(\theta) + H_{i2}(\theta),$$

$$H_{i1}(\theta) = d(x_i, L_i) \left\{ Y_i - \mathbb{E}[Y_{x_i, M_{x_i}} | L_i; \psi] \right\}.$$

Unbiasedness of the estimating equation (w.r.t. 0) - first component

$$\begin{aligned} \mathbb{E}[H_1(\theta) | X = x, L] &= d(x, L) \left\{ \mathbb{E}[Y | X = x, L] - \mathbb{E}[Y_{x, M_x} | L; \psi] \right\} \\ &= d(x, L) \left\{ \mathbb{E}[Y_x | X = x, L] - \mathbb{E}[Y_{x, M_x} | L; \psi] \right\} \\ &= d(x, L) \left\{ \mathbb{E}[Y_x | L] - \mathbb{E}[Y_{x, M_x} | L; \psi] \right\} \\ &= d(x, L) \left\{ \mathbb{E}[Y_{x, M_x} | L] - \mathbb{E}[Y_{x, M_x} | L; \psi] \right\} \\ &= 0. \end{aligned}$$

where, the second equation stems from the conditional exchangeability $Y_x \perp\!\!\!\perp X | L$ and the last equation stems id the model $\mathbb{E}[Y_{x, M_x} | L; \psi]$ is correctly specified.

Definition

$$H_i(\theta) = H_{i1}(\theta) + H_{i2}(\theta),$$

$$H_{i2}(\theta) = d(X_i, L_i) \left\{ \mathbb{E}[Y \mid X = 1 - X_i, M_i, L_i; \beta] - \mathbb{E}[Y_{(1-X_i), M_{X_i}} \mid L_i; \theta] \right\}.$$

Unbiasedness of the estimating equation (w.r.t. 0) - second component

$$\begin{aligned} \mathbb{E}[H_{i2}(\theta) \mid X = x, L] &= \mathbb{E}[\mathbb{E}[H_{i2}(\theta) \mid X = x, L, M] \mid X = x, L] \\ &= d(x, L) \left\{ \mathbb{E}[\mathbb{E}[Y \mid X = 1 - x, M, L; \beta] \mid X = x, L] - \mathbb{E}[Y_{(1-x), M_x} \mid L; \psi] \right\} \\ &= d(x, L) \left\{ \mathbb{E}[Y_{(1-x), M_x} \mid L; \beta] - \mathbb{E}[Y_{(1-x), M_x} \mid L; \psi] \right\} \\ &= 0, \end{aligned}$$

The second equality stems from the mediation formula, and the one before last inequality stems from correct models specification $\mathbb{E}[Y \mid X, M, L; \beta]$, and $\mathbb{E}[Y_{x, M_x} \mid L; \psi]$.

Natural Effects Model

$$\text{logit}\{\mathbb{P}(Y_{x,M_{x^*}} = 1 \mid L; \psi)\} = \psi_3 + \psi_0 x + \psi_1 x^* + \psi_2 L.$$

$$\text{NDE}(x^*, L) = \psi_0, \quad \text{NIE}(x, L) = \psi_1.$$

```
Natural effect model
with robust standard errors based on the sandwich estimator
---
Exposure: attbin
Mediator(s): negaffbin
---
Parameter estimates:
      Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.92025    0.68901  -1.336  0.18167
attbin0      0.51321    0.21752   2.359  0.01831 *
attbin1      0.22910    0.07007   3.270  0.00108 **
genderM      0.29136    0.22516   1.294  0.19567
educM        0.34194    0.48150   0.710  0.47760
educH        0.51189    0.48780   1.049  0.29400
age         -0.01219    0.01190  -1.025  0.30553
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

Imputation-based estimator

Structural mean model for the nested counterfactual

$$\text{logit}\left\{\mathbb{P}(Y_{x,M_{x^*}} = 1 \mid L; \psi)\right\} = \psi_3 + \psi_0 x + \psi_1 x^* + \psi_2 L,$$

$$P_M = \frac{\psi_1}{\psi_0 + \psi_1}$$

$$\hat{P}_M = \frac{0.22910}{0.22910 + 0.51321} = 0.3086336.$$

with a 95% CI

$$[0.07516409, 0.54210312]$$

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Structure of the weighting-based estimator

The parameters are

$$\theta = (\beta^T, \gamma^T)^T,$$

and the weighting-based estimator solves the system

$$\sum_{i=1}^n \begin{pmatrix} S_i(\beta) \\ H_i(\theta) \end{pmatrix} = 0,$$

where

$$H_i(\theta) = H_{i0}(\theta) + H_{i1}(\theta).$$

$$H_{i,x^*}(\theta) = d(X_i, L_i) \frac{\mathbb{P}(M_i | X = x^*, L_i; \gamma)}{\mathbb{P}(M_i | X_i, L_i; \gamma)} \left\{ Y_i - \mathbb{E}[Y_{X_i, M_{x^*}} | L_i; \psi] \right\}.$$

Here, $d(X_i, L_i)$ has the same dimension as ψ .

The estimating functions of the causal parameters are

$$H_i(\theta) = H_{i0}(\theta) + H_{i1}(\theta),$$

$$H_{i,x^*}(\theta) = d(X_i, L_i) \frac{\mathbb{P}(M_i | X = x^*, L_i; \gamma)}{\mathbb{P}(M_i | X_i, L_i; \gamma)} \left\{ Y_i - \mathbb{E}[Y_{X_i, M_{x^*}} | L_i; \psi] \right\}.$$

$$\begin{aligned} \mathbb{E}[H_{x^*}(\theta) | X = x, L] &= \mathbb{E}[\mathbb{E}[H_{x^*}(\theta) | X = x, M, L] \mid X = x, L] \\ &= d(x, L) \mathbb{E} \left[\frac{\mathbb{P}(M | X = x^*, L; \gamma)}{\mathbb{P}(M | X = x, L; \gamma)} \left\{ \mathbb{E}[Y | X = x, M, L] - \mathbb{E}[Y_{x, M_{x^*}} | L; \psi] \right\} \mid X = x, L \right] \\ &= d(x, L) \sum_m \left[\frac{\mathbb{P}(M = m | X = x^*, L; \gamma)}{\mathbb{P}(M = m | X = x, L; \gamma)} \left\{ \mathbb{E}[Y | X = x, M = m, L] - \mathbb{E}[Y_{x, M_{x^*}} | L; \psi] \right\} \right] \mathbb{P}(M = m | X = x, L) \\ &= d(x, L) \sum_m \left[\mathbb{P}(M = m | X = x^*, L; \gamma) \left\{ \mathbb{E}[Y | X = x, M = m, L] - \mathbb{E}[Y_{x, M_{x^*}} | L; \psi] \right\} \right] \end{aligned}$$

$$\begin{aligned}\mathbb{E}[H_{x^*}(\theta) \mid X = x, L] &= \mathbb{E}[\mathbb{E}[H_{x^*}(\theta) \mid X = x, M, L] \mid X = x, L] \\&: \\&= d(x, L) \sum_m \left[\mathbb{P}(M = m \mid X = x^*, L; \gamma) \left\{ \mathbb{E}[Y \mid X = x, M = m, L] - \mathbb{E}[Y_{x, M_{x^*}} \mid L; \psi] \right\} \right] \\&= d(x, L) [\mathbb{E}[\mathbb{E}[Y \mid X = x, M = m, L] \mid X = x^*, L; \gamma] - \mathbb{E}[Y_{x, M_{x^*}} \mid L; \psi]] \\&= d(x, L) [\mathbb{E}[Y_{x, M_{x^*}} \mid L; \gamma] - \mathbb{E}[Y_{x, M_{x^*}} \mid L; \psi]] \\&= 0,\end{aligned}$$

where the last three equations stem from

- Assuming a correct model for $\mathbb{P}(M \mid X, L; \gamma)$,
- The mediation formula
- Correct model for $\mathbb{E}[Y_{x, M_{x^*}} \mid L; \psi]$,

respectively.

Natural Effects Model

$$\text{logit}\{\mathbb{P}(Y_{x,M_{x^*}} = 1 \mid L; \psi)\} = \psi_3 + \psi_0 x + \psi_1 x^* + \psi_2 L.$$

$$\text{NDE}(x^*, L) = \psi_0, \quad \text{NIE}(x, L) = \psi_1.$$

```
Natural effect model
with robust standard errors based on the sandwich estimator
---
Exposure: attbin
Mediator(s): negaffbin
---
Parameter estimates:
      Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.952868   0.702083 -1.357 0.174717
attbin0      0.523624   0.218993  2.391 0.016800 *
attbin1      0.222259   0.067140  3.310 0.000932 ***
genderM      0.283234   0.230932  1.226 0.220017
educM        0.230908   0.492746  0.469 0.639345
educH        0.421673   0.499161  0.845 0.398243
age          -0.009679   0.012177 -0.795 0.426696
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Weighting-based estimator

Definition and basic properties

- Definition: The proportion mediated (P_M) is the fraction of the total effect of an exposure on an outcome that is explained by a given mediator. The P_M is defined as the ratio of the indirect to total effect, i.e.,

$$P_M = \frac{\text{Indirect Effect}}{\text{Total Effect}}.$$

- Interpretation:
 - P_M close to 1: nearly all of the exposure's effect on the outcome is transmitted via the mediator.
 - P_M near 0: the mediator accounts for little or none of the exposure's effect.
- Same direction requirement:
 - Typically used if total, direct, and indirect effects have **the same sign** (all positive or all negative).
 - If the mediator suppresses or reverses part of the effect, P_M can exceed 1 or become less intuitive (i.e., not really represents a proportion).
- Indirect and total effects size matters:
 - Even large values of P_M might correspond to a small overall effect if the total effect itself is small.
 - Conversely, a moderate P_M can represent a significant absolute change if the total effect is large.

Structural mean model for the nested counterfactual

$$\text{logit}\left\{\mathbb{P}(Y_{x,M_x^*} = 1 \mid L; \psi)\right\} = \psi_3 + \psi_0 x + \psi_1 x^* + \psi_2 L,$$

$$P_M = \frac{\psi_1}{\psi_0 + \psi_1}$$

$$\hat{P}_M = \frac{0.222259}{0.222259 + 0.523624} = 0.2979816.$$

with a 95% CI

$$[0.0723026, 0.5236606]$$

The two estimator (imputation-based and weighting-based) are practically coincides with each other.

- 1 Motivating example
- 2 Controlled direct effect
- 3 Natural direct and indirect effects - recap
- 4 Identification and Estimation
 - Imputation-based estimator
 - Weighting-based estimator
- 5 References

References

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